

cv_validated_voters_ensemble

scalars_cv_validated_voters_ensemble

key	value
n_samples	10482
main_kwargs_applied.substitute	False
main_kwargs_applied.normalize_final_letters	True
timings_seconds.cross_wer	12.6
timings_seconds.ranking	151.2
timings_seconds.model_filtration	929.6
timings_seconds.rank_distribution	1193.4
ranking_stats.n_models	7
ranking_stats.pool	seamless_m4t+whisper_v3+fastconformer+w2vbert+hubert+wav2vec_sb+cohere
rank_distribution_summary.n_subpools	64
rank_distribution_summary.n_boot_inner	1000
rank_distribution_summary.n_fusions	99
rank_distribution_summary.n_scorings	294
rank_distribution_summary.seconds_fuse_and_score	255.3
rank_distribution_summary.seconds_bootstrap	8.4
rank_distribution_summary.weighting	filtration
rank_distribution_summary.exponent	1
rank_distribution_summary.mean_entropy	0.2348
rank_distribution_summary.max_entropy	0.426
rank_distribution_summary.n_models_modal_differs_from_full	0

ranking_cv_validated_voters_ensemble

model	role	n_samples	wer_pseudo_loo	cer_pseudo_loo	rank_pseudo
cohere	voter	10482	5.45	1.5811	1
fastconformer	voter	10482	6.8236	2.0189	2
hubert	voter	10482	9.0172	2.4016	3
seamless_m4t	voter	10482	9.6853	3.3181	4
whisper_v3	voter	10482	13.0163	4.1725	5
w2vbert	voter	10482	22.5822	6.1957	6
wav2vec_sb	voter	10482	27.8457	9.2243	7

model_confidence_cv_validated_voters_ensemble

model	rank_pseudo_full	rank_modal	p_modal	p_at_full_rank	entropy_norm	rank_mean	rank_sd	rank_ci_lo	rank_ci_hi	agrees_with_full
cohere	1	1	0.808	0.808	0.3203	1.2515	0.5759	1	3	True
fastconformer	2	2	0.7213	0.7213	0.426	1.9449	0.5915	1	3	True
hubert	3	3	0.7204	0.7204	0.4202	3.0523	0.5612	2	4	True
seamless_m4t	4	4	0.7565	0.7565	0.3904	3.7911	0.5642	2	5	True
whisper_v3	5	5	0.963	0.963	0.0864	4.9602	0.209	4	5	True
w2vbert	6	6	1	1	-0	6	0	6	6	True
wav2vec_sb	7	7	1	1	-0	7	0	7	7	True

rank_distribution_cv_validated_voters_ensemble

model	rank_1	rank_2	rank_3	rank_4	rank_5	rank_6	rank_7
seamless_m4t	0	0.0395	0.167	0.7565	0.037	0	0
whisper_v3	0	0	0.0027	0.0343	0.963	0	0
fastconformer	0.1793	0.7213	0.0747	0.0247	0	0	0
w2vbert	0	0	0	0	0	1	0
hubert	0.0127	0.0946	0.7204	0.1723	0	0	0
wav2vec_sb	0	0	0	0	0	0	1
cohere	0.808	0.1446	0.0352	0.0122	0	0	0

pair_probability_cv_validated_voters_ensemble

pair	pred_winner_full	gap_wer_pseudo	p_first_beats_second	p_supports_full
fastconformer cohere	cohere	1.3736	0.1793	0.8207
seamless_m4t hubert	hubert	0.6681	0.1725	0.8275
fastconformer hubert	fastconformer	2.1936	0.9144	0.9144
seamless_m4t fastconformer	fastconformer	2.8617	0.0385	0.9615
hubert cohere	cohere	3.5672	0.0374	0.9626
seamless_m4t whisper_v3	seamless_m4t	3.331	0.963	0.963
seamless_m4t cohere	cohere	4.2353	0.0349	0.9651
whisper_v3 hubert	hubert	3.9991	0.0027	0.9973
hubert wav2vec_sb	hubert	18.8285	1	1
w2vbert cohere	cohere	17.1322	0	1
w2vbert wav2vec_sb	w2vbert	5.2635	1	1
w2vbert hubert	hubert	13.565	0	1
fastconformer wav2vec_sb	fastconformer	21.0221	1	1
whisper_v3 cohere	cohere	7.5663	0	1
whisper_v3 wav2vec_sb	whisper_v3	14.8294	1	1
whisper_v3 w2vbert	whisper_v3	9.5659	1	1
whisper_v3 fastconformer	fastconformer	6.1927	0	1

pair	pred_winner_full	gap_wer_pseudo	p_first_beats_second	p_supports_full
seamless_m4t wav2vec_sb	seamless_m4t	18.1604	1	1
seamless_m4t w2vbert	seamless_m4t	12.8968	1	1
fastconformer w2vbert	fastconformer	15.7586	1	1
wav2vec_sb cohere	cohere	22.3957	0	1

model_filtration_cv_validated_voters_ensemble

model	n_subpools	f_bar	f_sd	f_min	f_max	survival
cohere	42	0.0412	0.0082	0.0301	0.059	0.9588
fastconformer	42	0.0666	0.0164	0.0454	0.1068	0.9334
hubert	42	0.0981	0.0254	0.0654	0.1606	0.9019
seamless_m4t	42	0.1261	0.0265	0.0936	0.2083	0.8739
whisper_v3	42	0.1911	0.0411	0.1177	0.2938	0.8089
w2vbert	42	0.4128	0.0588	0.3189	0.5225	0.5872
wav2vec_sb	42	0.5179	0.04	0.4436	0.5967	0.4821

by_pool_size_cv_validated_voters_ensemble

k	pools	total_weight	mean_entropy	max_entropy
4	35	0.4762	0.0217	0.0982
5	21	0.3571	0.0272	0.1587
6	7	0.1429	0.0154	0.0807
7	1	0.0238	0.0012	0.0012

cross_wer_cv_validated_voters_ensemble

	seamless_m4t	whisper_v3	fastconformer	w2vbert	hubert	wav2vec_sb	cohere
seamless_m4t	0	15.72	11.73	25.68	13.77	30.64	10.12
whisper_v3	15.72	0	14.93	26.19	17.02	31.69	13.81
fastconformer	11.73	14.93	0	24.05	10.29	28.79	7.03
w2vbert	25.68	26.19	24.05	0	25.3	37.18	23.29
hubert	13.77	17.02	10.29	25.3	0	29.57	9.2
wav2vec_sb	30.64	31.69	28.79	37.18	29.57	0	28.78
cohere	10.12	13.81	7.03	23.29	9.2	28.78	0

subpool_filtration_cv_validated_voters_ensemble (first 60 of 294 rows)

pool	k	model	n_samples	filtered_out	filtered_out_rate
seamless_m4t+whisper_v3+fastconformer+w2vbert	4	seamless_m4t	10482	1304	0.1244
seamless_m4t+whisper_v3+fastconformer+w2vbert	4	whisper_v3	10482	1859	0.1774
seamless_m4t+whisper_v3+fastconformer+w2vbert	4	fastconformer	10482	878	0.0838
seamless_m4t+whisper_v3+fastconformer+w2vbert	4	w2vbert	10482	4670	0.4455
seamless_m4t+whisper_v3+fastconformer+hubert	4	seamless_m4t	10482	1632	0.1557
seamless_m4t+whisper_v3+fastconformer+hubert	4	whisper_v3	10482	2622	0.2501
seamless_m4t+whisper_v3+fastconformer+hubert	4	fastconformer	10482	947	0.0903
seamless_m4t+whisper_v3+fastconformer+hubert	4	hubert	10482	1460	0.1393
seamless_m4t+whisper_v3+fastconformer+wav2vec_sb	4	seamless_m4t	10482	1047	0.0999
seamless_m4t+whisper_v3+fastconformer+wav2vec_sb	4	whisper_v3	10482	1724	0.1645
seamless_m4t+whisper_v3+fastconformer+wav2vec_sb	4	fastconformer	10482	657	0.0627
seamless_m4t+whisper_v3+fastconformer+wav2vec_sb	4	wav2vec_sb	10482	5519	0.5265
seamless_m4t+whisper_v3+fastconformer+cohere	4	seamless_m4t	10482	1693	0.1615
seamless_m4t+whisper_v3+fastconformer+cohere	4	whisper_v3	10482	2788	0.266
seamless_m4t+whisper_v3+fastconformer+cohere	4	fastconformer	10482	1047	0.0999
seamless_m4t+whisper_v3+fastconformer+cohere	4	cohere	10482	566	0.054
seamless_m4t+whisper_v3+w2vbert+hubert	4	seamless_m4t	10482	1265	0.1207
seamless_m4t+whisper_v3+w2vbert+hubert	4	whisper_v3	10482	1868	0.1782
seamless_m4t+whisper_v3+w2vbert+hubert	4	w2vbert	10482	4547	0.4338
seamless_m4t+whisper_v3+w2vbert+hubert	4	hubert	10482	1234	0.1177
seamless_m4t+whisper_v3+w2vbert+wav2vec_sb	4	seamless_m4t	10482	985	0.094
seamless_m4t+whisper_v3+w2vbert+wav2vec_sb	4	whisper_v3	10482	1234	0.1177
seamless_m4t+whisper_v3+w2vbert+wav2vec_sb	4	w2vbert	10482	3343	0.3189
seamless_m4t+whisper_v3+w2vbert+wav2vec_sb	4	wav2vec_sb	10482	4665	0.445
seamless_m4t+whisper_v3+w2vbert+cohere	4	seamless_m4t	10482	1284	0.1225
seamless_m4t+whisper_v3+w2vbert+cohere	4	whisper_v3	10482	1953	0.1863
seamless_m4t+whisper_v3+w2vbert+cohere	4	w2vbert	10482	4787	0.4567
seamless_m4t+whisper_v3+w2vbert+cohere	4	cohere	10482	593	0.0566
seamless_m4t+whisper_v3+hubert+wav2vec_sb	4	seamless_m4t	10482	1040	0.0992
seamless_m4t+whisper_v3+hubert+wav2vec_sb	4	whisper_v3	10482	1695	0.1617
seamless_m4t+whisper_v3+hubert+wav2vec_sb	4	hubert	10482	928	0.0885
seamless_m4t+whisper_v3+hubert+wav2vec_sb	4	wav2vec_sb	10482	5395	0.5147
seamless_m4t+whisper_v3+hubert+cohere	4	seamless_m4t	10482	1657	0.1581
seamless_m4t+whisper_v3+hubert+cohere	4	whisper_v3	10482	2755	0.2628
seamless_m4t+whisper_v3+hubert+cohere	4	hubert	10482	1579	0.1506
seamless_m4t+whisper_v3+hubert+cohere	4	cohere	10482	559	0.0533
seamless_m4t+whisper_v3+wav2vec_sb+cohere	4	seamless_m4t	10482	1032	0.0985
seamless_m4t+whisper_v3+wav2vec_sb+cohere	4	whisper_v3	10482	1758	0.1677
seamless_m4t+whisper_v3+wav2vec_sb+cohere	4	wav2vec_sb	10482	5612	0.5354
seamless_m4t+whisper_v3+wav2vec_sb+cohere	4	cohere	10482	433	0.0413

pool	k	model	n_samples	filtered_out	filtered_out_rate
seamless_m4t+fastconformer+w2vbert+hubert	4	seamless_m4t	10482	1520	0.145
seamless_m4t+fastconformer+w2vbert+hubert	4	fastconformer	10482	754	0.0719
seamless_m4t+fastconformer+w2vbert+hubert	4	w2vbert	10482	5044	0.4812
seamless_m4t+fastconformer+w2vbert+hubert	4	hubert	10482	1093	0.1043
seamless_m4t+fastconformer+w2vbert+wav2vec_sb	4	seamless_m4t	10482	1018	0.0971
seamless_m4t+fastconformer+w2vbert+wav2vec_sb	4	fastconformer	10482	544	0.0519
seamless_m4t+fastconformer+w2vbert+wav2vec_sb	4	w2vbert	10482	3705	0.3535
seamless_m4t+fastconformer+w2vbert+wav2vec_sb	4	wav2vec_sb	10482	4887	0.4662
seamless_m4t+fastconformer+w2vbert+cohere	4	seamless_m4t	10482	1559	0.1487
seamless_m4t+fastconformer+w2vbert+cohere	4	fastconformer	10482	795	0.0758
seamless_m4t+fastconformer+w2vbert+cohere	4	w2vbert	10482	5244	0.5003
seamless_m4t+fastconformer+w2vbert+cohere	4	cohere	10482	427	0.0407
seamless_m4t+fastconformer+hubert+wav2vec_sb	4	seamless_m4t	10482	1334	0.1273
seamless_m4t+fastconformer+hubert+wav2vec_sb	4	fastconformer	10482	561	0.0535
seamless_m4t+fastconformer+hubert+wav2vec_sb	4	hubert	10482	825	0.0787
seamless_m4t+fastconformer+hubert+wav2vec_sb	4	wav2vec_sb	10482	5781	0.5515
seamless_m4t+fastconformer+hubert+cohere	4	seamless_m4t	10482	2183	0.2083
seamless_m4t+fastconformer+hubert+cohere	4	fastconformer	10482	1109	0.1058
seamless_m4t+fastconformer+hubert+cohere	4	hubert	10482	1683	0.1606
seamless_m4t+fastconformer+hubert+cohere	4	cohere	10482	571	0.0545

subpool_ranks_cv_validated_voters_ensemble (first 60 of 520 rows)

pool	k	model	rank	count
seamless_m4t+whisper_v3+fastconformer+w2vbert	4	seamless_m4t	4	1000
seamless_m4t+whisper_v3+fastconformer+w2vbert	4	whisper_v3	5	1000
seamless_m4t+whisper_v3+fastconformer+w2vbert	4	fastconformer	2	5
seamless_m4t+whisper_v3+fastconformer+w2vbert	4	fastconformer	3	995
seamless_m4t+whisper_v3+fastconformer+w2vbert	4	w2vbert	6	1000
seamless_m4t+whisper_v3+fastconformer+w2vbert	4	hubert	2	995
seamless_m4t+whisper_v3+fastconformer+w2vbert	4	hubert	3	5
seamless_m4t+whisper_v3+fastconformer+w2vbert	4	wav2vec_sb	7	1000
seamless_m4t+whisper_v3+fastconformer+w2vbert	4	cohere	1	1000
seamless_m4t+whisper_v3+fastconformer+hubert	4	seamless_m4t	3	173
seamless_m4t+whisper_v3+fastconformer+hubert	4	seamless_m4t	4	827
seamless_m4t+whisper_v3+fastconformer+hubert	4	whisper_v3	5	1000
seamless_m4t+whisper_v3+fastconformer+hubert	4	fastconformer	2	1000
seamless_m4t+whisper_v3+fastconformer+hubert	4	w2vbert	6	1000
seamless_m4t+whisper_v3+fastconformer+hubert	4	hubert	3	827

pool	k	model	rank	count
seamless_m4t+whisper_v3+fastconformer+hubert	4	hubert	4	173
seamless_m4t+whisper_v3+fastconformer+hubert	4	wav2vec_sb	7	1000
seamless_m4t+whisper_v3+fastconformer+hubert	4	cohere	1	1000
seamless_m4t+whisper_v3+fastconformer+wav2vec_sb	4	seamless_m4t	4	1000
seamless_m4t+whisper_v3+fastconformer+wav2vec_sb	4	whisper_v3	5	1000
seamless_m4t+whisper_v3+fastconformer+wav2vec_sb	4	fastconformer	2	26
seamless_m4t+whisper_v3+fastconformer+wav2vec_sb	4	fastconformer	3	974
seamless_m4t+whisper_v3+fastconformer+wav2vec_sb	4	w2vbert	6	1000
seamless_m4t+whisper_v3+fastconformer+wav2vec_sb	4	hubert	2	974
seamless_m4t+whisper_v3+fastconformer+wav2vec_sb	4	hubert	3	26
seamless_m4t+whisper_v3+fastconformer+wav2vec_sb	4	wav2vec_sb	7	1000
seamless_m4t+whisper_v3+fastconformer+wav2vec_sb	4	cohere	1	1000
seamless_m4t+whisper_v3+fastconformer+cohere	4	seamless_m4t	4	1000
seamless_m4t+whisper_v3+fastconformer+cohere	4	whisper_v3	5	1000
seamless_m4t+whisper_v3+fastconformer+cohere	4	fastconformer	2	1000
seamless_m4t+whisper_v3+fastconformer+cohere	4	w2vbert	6	1000
seamless_m4t+whisper_v3+fastconformer+cohere	4	hubert	3	1000
seamless_m4t+whisper_v3+fastconformer+cohere	4	wav2vec_sb	7	1000
seamless_m4t+whisper_v3+fastconformer+cohere	4	cohere	1	1000
seamless_m4t+whisper_v3+w2vbert+hubert	4	seamless_m4t	3	886
seamless_m4t+whisper_v3+w2vbert+hubert	4	seamless_m4t	4	114
seamless_m4t+whisper_v3+w2vbert+hubert	4	whisper_v3	5	1000
seamless_m4t+whisper_v3+w2vbert+hubert	4	fastconformer	2	1000
seamless_m4t+whisper_v3+w2vbert+hubert	4	w2vbert	6	1000
seamless_m4t+whisper_v3+w2vbert+hubert	4	hubert	3	114
seamless_m4t+whisper_v3+w2vbert+hubert	4	hubert	4	886
seamless_m4t+whisper_v3+w2vbert+hubert	4	wav2vec_sb	7	1000
seamless_m4t+whisper_v3+w2vbert+hubert	4	cohere	1	1000
seamless_m4t+whisper_v3+w2vbert+wav2vec_sb	4	seamless_m4t	4	994
seamless_m4t+whisper_v3+w2vbert+wav2vec_sb	4	seamless_m4t	5	6
seamless_m4t+whisper_v3+w2vbert+wav2vec_sb	4	whisper_v3	4	6
seamless_m4t+whisper_v3+w2vbert+wav2vec_sb	4	whisper_v3	5	994
seamless_m4t+whisper_v3+w2vbert+wav2vec_sb	4	fastconformer	2	1000
seamless_m4t+whisper_v3+w2vbert+wav2vec_sb	4	w2vbert	6	1000
seamless_m4t+whisper_v3+w2vbert+wav2vec_sb	4	hubert	3	1000
seamless_m4t+whisper_v3+w2vbert+wav2vec_sb	4	wav2vec_sb	7	1000
seamless_m4t+whisper_v3+w2vbert+wav2vec_sb	4	cohere	1	1000
seamless_m4t+whisper_v3+w2vbert+cohere	4	seamless_m4t	4	1000
seamless_m4t+whisper_v3+w2vbert+cohere	4	whisper_v3	5	1000
seamless_m4t+whisper_v3+w2vbert+cohere	4	fastconformer	1	1000

pool	k	model	rank	count
seamless_m4t+whisper_v3+w2vbert+cohere	4	w2vbert	6	1000
seamless_m4t+whisper_v3+w2vbert+cohere	4	hubert	3	1000
seamless_m4t+whisper_v3+w2vbert+cohere	4	wav2vec_sb	7	1000
seamless_m4t+whisper_v3+w2vbert+cohere	4	cohere	2	1000
seamless_m4t+whisper_v3+hubert+wav2vec_sb	4	seamless_m4t	3	31